



Technical Data Sheet

- Extra Performance
- Meets IEC 60296 - Higher Oxidation Stability & Low Sulphur Content

Diala S4 ZX-I

Premium Inhibited Electrical Insulating Oil

Shell Diala S4 ZX-I is the new electrical insulating oil from Shell designed to meet the challenges presented by the latest power transformers. It offers an extended oil life with the peace of mind of zero sulphur content.

Shell Diala S4 ZX-I is manufactured from zero sulphur base oils produced using Shell's GTL (gas-to-liquid) technology. These base oils offer a high degree of compositional consistency and have an excellent response to anti-oxidant. In addition they are globally available and free from PCBs, DBDS and passivators.

Shell Diala S4 ZX-I meets both the established and new industry copper corrosion tests.

DESIGNED TO MEET CHALLENGES

Performance, Features & Benefits

Extended oil life

Shell Diala S4 ZX-I is a fully inhibited oil giving outstanding oxidation performance and an extended oil life. Shell Diala S4 ZX-I is also suitable for use in highly loaded applications.

Transformer protection

Shell Diala S4 ZX-I is manufactured from a zero sulphur* base oil, making it intrinsically non-corrosive towards copper, without the need for passivation or other additives.

Shell Diala S4 ZX-I meets all relevant tests for copper corrosion, namely the established DIN 51353 (Silver Strip Test), ASTM D1275, and also the latest more severe tests: IEC 62535 and ASTM D1275B.

*Sulphur content below 1ppm detection limit of ASTM D5185

System efficiency

The good low temperature viscometric properties of the oil ensure proper heat transfer inside the transformer, even from very low starting temperatures.

Shell Diala S4 ZX-I is specially dried and handled to achieve a low water content and retain a high breakdown voltage at point of delivery. This enables it to be used in many applications without further treatment.

Main Applications



Specifications, Approvals & Recommendations

- IEC 60296 (2012): Table 2 Transformer Oil (I) (Inhibited oil) Section 7.1 ("Higher oxidation stability & low sulphur content")
For a full listing of equipment approvals and recommendations, please consult your local Shell Technical Help Desk, or the OEM Approvals website.

Typical Physical Characteristics

Properties			Method	IEC 60296 Table 2 + section 7.1	Diala S4 ZX-I Typical
Appearance			IEC 60296	Clear, free from sediment and suspended matter	Complies
Density	@20°C	kg/m ³	ISO 3675	Max. 895	805
Kinematic Viscosity	@40°C	mm ² /s	ISO 3104	Max. 12	9.6
Kinematic Viscosity	@-30°C	mm ² /s	ISO 3104	Max. 1 800	382
Flashpoint P.M.		°C	ISO 2719	Min. 135	191
Pour Point		°C	ISO 3016	Max. -40	-42
Neutralisation value	mg KOH/g		IEC 62021-1	Max. 0.01	<0.01
Total Sulphur Content	mg/kg		ASTM D5185	Max. 500	<1
Corrosive Sulphur			DIN 51353	Not corrosive	Not corrosive

Properties		Method	IEC 60296 Table 2 + section 7.1	Diala S4 ZX-I Typical
Potentially Corrosive Sulphur		IEC 62535	Not corrosive	Not corrosive
Corrosive Sulphur		ASTM D1275 B		Not corrosive
Breakdown Voltage Untreated	kV	IEC 60156	Min. 30	60
Breakdown Voltage After Treatment	kV	IEC 60156	Min. 70	75
Dielectric Dissipation Factor	@90°C DDF	IEC 60247	Max 0.005	<0.001
Oxidation Stability	500h / 120°C	IEC 61125 C	Section 7.1 Limits	
Total Acidity	mg KOH/g %m	IEC 61125 C	Max 0.3	0.02
Sludge	mg KOH/g %m	IEC 61125 C	Max 0.05	<0.01
Dielectric Dissipation Factor (DDF @90°C)	mg KOH/g %m	IEC 61125 C	Max 0.05	0.001
Water content (drums and IBC)	mg/kg	IEC 60814	Max 40	6
Water content (Bulk)	mg/kg	IEC 60814	Max 30	6
2-Furfural and related compounds content	mg/kg	IEC 61198	Not detectable	Complies
Metal passivator additives	mg/kg	IEC 60666	Not detectable	Complies
Oxidation inhibitor content	% mass			0.2
PCB content	mg/kg	IEC 61619	Not detectable	Complies

These characteristics are typical of current production. Whilst future production will conform to Shell's specification, variations in these characteristics may occur.

Health, Safety & Environment

■ Health and Safety

Shell Diala S4 ZX-I is unlikely to present any significant health or safety hazard when properly used in the recommended application and good standards of personal hygiene are maintained.

Shell Diala S4 ZX-I is free from polychlorinated biphenyls (PCB).

Avoid contact with skin. Use impervious gloves with used oil. After skin contact, wash immediately with soap and water.

Guidance on Health and Safety is available on the appropriate Material Safety Data Sheet, which can be obtained from your Shell representative.

■ Protect the Environment

Take used oil to an authorised collection point. Do not discharge into drains, soil or water.

Additional Information

Storage precautions

The critical electrical properties of Shell Diala are easily compromised by trace contamination with foreign material. Typically encountered contaminants include moisture, particles, fibres and surfactants. Therefore, it is imperative that electrical insulating oils be kept clean and dry.

It is strongly recommended that storage containers be dedicated for electrical service and include air-tight seals. It is further recommended that electrical insulating oils are stored indoors in climate-controlled environments.

Advice

Advice on applications not covered here may be obtained from your Shell representative.